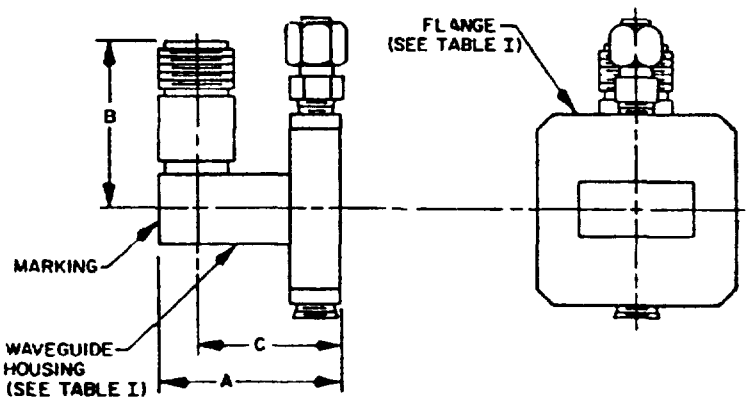
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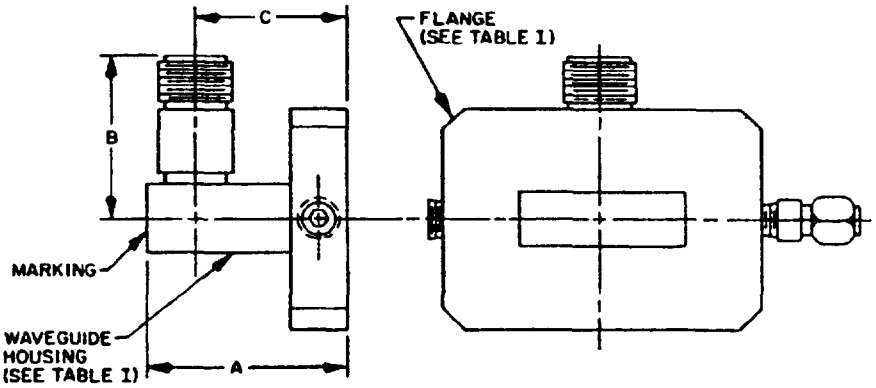
ADAPTERS, COAXIAL TO WAVEGUIDE
(SERIES N TO WAVEGUIDE - CLASS I), PRESSURIZABLE

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the adapters described herein shall consist of this document and the latest issue of Specification MIL-A-22641.



Configuration 1. Configuration for adapters with the pressure fitting in the broad wall for dash numbers -001 thru -020.



Configuration 2 Configuration for adapters with the pressure fitting in the narrow wall for dash numbers -020 thru -040

FIGURE 1. Adapters

FSC 5985

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TABLE I. Adapters, coaxial to waveguide (class I - Series N to waveguide)

Dash number	Frequency range (GHz)	VSWR over entire and over 50% of frequency range 1/ 2/ (max)	Insertion loss dB (max) 2/	Waveguide housing 3/	Flange and mounting method 4/	Config-uration	Dimensions A (max)	Dimensions B (max)	Dimensions C (max)	Waveguide and flange material	Mating flange
001	2.3-4.4	1.15 1 and 1 08 1	0 05	M85/4-001 RG-109/U	M3922/71-13 (Contact) Butt	1	3.13 (79.50)	3.40 (86.36)	2 13 (54.10)	Copper alloy	UG-509/U RE-F2687361
002				M85/4-001 RG-109/U	M3922/71-01 (Contact) Socket					Copper alloy	UG-509/U RE-F2687361
003				M85/4-002	M3922/71-14 (Contact) Butt					Aluminum alloy	- - -
004	2 3-4.4			M85/4-002	M3922/71-02 (Contact) Socket		3.13 (79.50)	3 40 (86.36)	2.13 (54.10)	Aluminum alloy	- - -
005	4.3-7.35			M85/1-061 RG-343	M3922/71-17 (Contact) Butt		2.00 (50.80)	1.46 (37.08)	1.60 (40.64)	Copper alloy	M3922/52-037
006				M85/1-061 RG-343	M3922/71-05 (Contact) Socket					Copper alloy	M3922/52-037
007				M85/1-060	M3922/71-18 (Contact) Butt					Aluminum alloy	M3922/52-038
008	4 3-7.35	1.15·1 and 1.08·1	0.05	M85/1-060	M3922/71-06 (Contact) Socket		2.00 (50.80)	1.46 (37.08)	1.60 (40.64)	Aluminum alloy	M3922/52-038
009	5.0-10 75	1 20 1 and 1.15 1	0 08	M85/4-003 RG-110/U	M3922/71-15 (Contact) Butt	1	1 50 (38.10)	1.75 (44.45)	1.38 (35.05)	Copper alloy	RE-F2687362 UG-511/U

See footnotes at end of table.

FIGURE 1 Adapters - Continued

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TABLE I. Adapters, coaxial to waveguide (class I - Series N to waveguide) - Continued.

Dash number	Frequency range (GHz)	VSWR over entire and over 50% of frequency range 1/ 2/ (max)	Insertion loss dB (max) 2/	Waveguide housing 3/	Flange and mounting method 4/	Configuration	Dimensions A (max)	Dimensions B (max)	Dimensions C (max)	Waveguide and flange material	Mating flange
010	5.0-10.75	1.20:1 and 1.15:1	0.08	M85/4-003 RG-110/U	M3922/71-03 (Contact) Socket	1	1.50 (38.10)	1.75 (44.45)	1.38 (35.05)	Copper alloy	UG-511/U RE-F2687362
011	↓	↓	↓	M85/4-004	M3922/71-16 (Contact) Butt	↓	↓	↓	↓	Aluminum alloy	---
012	5.0-10.75	↓	↓	M85/4-004	M3922/71-04 (Contact) Socket	↓	1.50 (38.10)	1.75 (44.45)	1.38 (35.05)	Aluminum alloy	---
013	7-11	↓	↓	M85/1-155 RG-320/U	M3922/72-05 (Cover) Butt	↓	2.00 (50.80)	1.63 (41.40)	1.63 (41.40)	Copper alloy	M3922/69-001
014	↓	↓	↓	↓	M3922/72-01 (Cover) Socket	↓	↓	↓	↓	↓	M3922/69-001
015	↓	↓	↓	↓	M3922/73-05 (Choke) Butt	↓	↓	↓	↓	↓	M3922/70-001, -013
016	↓	↓	↓	M85/1-155 RG-320/U	M3922/73-01 (Choke) Socket	↓	↓	↓	↓	Copper alloy	M3922/70-001, -013
017	↓	↓	↓	M85/1-158	M3922/72-06 (Cover) Butt	↓	↓	↓	↓	Aluminum alloy	M3922/69-002
018	7-11	1.20:1 and 1.15:1	0.08	M85/1-158	M3922/72-02 (Cover) Socket	1	2.00 (50.80)	1.63 (41.40)	1.63 (41.40)	Aluminum alloy	M3922/69-002

See footnotes at end of table.

FIGURE 1. Adapters - Continued.

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TABLE I Adapters, coaxial to waveguide (class I - Series N to waveguide) - Continued

Dash number	Frequency range (GHz)	VSWR over entire and over 50% of frequency range 1/ 2/ (max)	Insertion loss dB (max) 2/	Waveguide housing 3/	Flange and mounting method 4/	Configuration	Dimensions A (max)	Dimensions B (max)	Dimensions C (max)	Waveguide and flange material	Mating flange
019	7-11	1.20 1 and 1.15 1	0.08	M85/1-158	M3922/73-06 (Choke) Butt	1	2.00 (50 80)	1 63 (41 40)	1.63 (41 40)	Aluminum alloy	M3922/52-038
020	7-11	1 20 1 and 1 15 1	0.08	M85/1-158	M3922/73-02 (Choke) Socket	1	2.00 (50.80)	1 63 (41 40)	1.63 (41 40)	Aluminum alloy	---
021	2.3-4.4	1.15·1 and 1.08 1	0.05	M85/4-001 RG-109/U	M3922/71-19 (Contact) Butt	2	3.13 (79 50)	3 40 (86.36)	2 13 (54 10)	Copper alloy	UG-509/U RE-F2687361
022				M85/4-001 RG-109/U	M3922/71-07 (Contact) Socket					Copper alloy	UG-509/U RE-F2687361
023				M85/4-002	M3922/71-20 (Contact) Butt					Aluminum alloy	---
024	2 3-4.4			M85/4-002	M3922/71-08 (Contact) Socket		3 13 (79.50)	3 40 (86 36)	2 13 (54.10)	Aluminum alloy	---
025	4 3-7 35			M85/1-061 RG-343	M3922/71-23 (Contact) Butt		2 20 (55 88)	1 80 (45 72)	1 70 (43.18)	Copper alloy	M3922/52-037
026				M85/1-061 RG-343	M3922/71-11 (Contact) Socket					Copper alloy	M3922/52-037
027	4.3-7.35	1.15 1 and 1.08·1	0.05	M85/1-060	M3922/71-24 (Contact) Butt	2	2.20 (55.88)	1 80 (45.72)	1 70 (43.18)	Aluminum alloy	M3922/52-038

See footnotes at end of table.

FIGURE 1 Adapters - Continued

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TABLE I. Adapters, coaxial to waveguide (class I - Series N to waveguide) - Continued.

Dash number	Frequency range (GHz)	VSWR over entire and over 50% of frequency range 1/ 2/ (max)	Insertion loss dB (max) 2/	Waveguide housing 3/	Flange and mounting method 4/	Config-uration	Dimensions A (max)	Dimensions B (max)	Dimensions C (max)	Waveguide and flange material	Mating Flange
028	4.3-7.35	1.15:1 and 1.08:1	0.05	M85/1-060	M3922/71-12 (Contact) Socket	2	2.20 (55.88)	1.80 (45.72)	1.70 (43.18)	Aluminum alloy	M3922/52-038
029	5.0-10.75	1.20:1 and 1.15:1	0.08	M85/4-003 RG-110/U	M3922/71-21 (Contact) Butt		1.50 (38.10)	1.75 (44.45)	1.38 (35.05)	Copper alloy	UG-511/U RE-F2687362
030				M85/4-003 RG-110/U	M3922/71-09 (Contact) Socket					Copper alloy	UG-511/U RE-F2687362
031				M85/4-004	M3922/71-22 (Contact) Butt					Aluminum alloy	---
032	5.0-10.75			M85/4-004	M3922/71-10 (Contact) Socket		1.50 (38.10)	1.75 (44.45)	1.38 (35.05)	Aluminum alloy	---
033	7-11			M85/1-155 RG-320/U	M3922/72-07 (Cover) Butt		2.00 (50.80)	1.63 (41.40)	1.63 (41.40)	Copper alloy	M3922/69-001
034					M3922/72-03 (Cover) Socket						M3922/69-001
035					M3922/73-07 (Choke) Butt						M3922/70-001, 013
036	7-11	1.20:1 and 1.15:1	0.08	M85/1-155 RG-320/U	M3922/73-03 (Choke) Socket	2	2.00 (50.80)	1.63 (41.40)	1.63 (41.40)	Copper alloy	M3922/70-001, 013

See footnotes at end of table.

FIGURE 1 Adapters - Continued

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TABLE I Adapters, coaxial to waveguide (class I - Series N to waveguide) - Continued

Dash number	Frequency range (GHz)	VSWR over entire and over 50% of frequency range 1/ 2/ (max)	Insertion loss dB (max) 2/	Waveguide housing 3/	Flange and mounting method 4/	Config-uration	Dimensions A (max)	Dimensions B (max)	Dimensions C (max)	Waveguide and flange material	Mating flange
037	7-11	1.20 1 and 1.15 1	0.08	M85/1-158	M3922/72-08 (Cover) Butt	2	2.00 (50.80)	1.63 (41.40)	1 63 (41.40)	Aluminum alloy	M3922/69-001, -002
038					M3922/72-04 (Cover) Socket						M3922/69-001, -002
039					M3922/73-08 (Choke) Butt						M3922/70-002, -014
040	7-11	1 20·1 and 1 15·1	0 08	M85/1-158	M3922/73-04 (Choke) Socket	2	2.00 (50.80)	1.63 (41.40)	1.63 (41 40)	Aluminum alloy	M3022/70-002, -014

1/ A larger attenuator pad may be used for VSWR test setup (see figure 1 of MIL-A-22641C) when measuring values less than 1.25:1.
2/ Swept frequency measuring techniques is an acceptable test method for determining VSWR and insertion loss values
3/ Waveguide in accordance with MIL-W-85.
4/ Flange in accordance with MIL-F-3922.

NOTES
1 Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3 Metric equivalents are in parentheses.

FIGURE 1. Adapters - Continued.

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REQUIREMENTS

Design, construction, and physical dimensions: See configurations 1 and 2

Operating temperature range: -54° to +125°C.

Frequency range See table I.

VSWR: See table I.

Insertion loss: See table I.

Flange, waveguide: See table I.

Housing, waveguide. See table I.

Finish for copper base alloy material: All exterior surfaces of copper base alloy material adapters, except the mating surface, shall be finished in accordance with finish number P213, type I, with the final film E of MIL-F-14072 conforming to type III, class 2, of MIL-E-15090

Pressurization With the flange end properly sealed, adapters shall be subjected to 30 lb_f/in^2 gage of dry nitrogen gas for a period of 5 minutes while immersed in tap water at a temperature of approximately 20°C During the immersion, any visible escaping bubbles from any surface of the adapter will be considered evidence of inadequate sealing

Connector The female connector shall mate and be compatible with the series N type male connector per MIL-C-39012

Marking See configurations 1 and 2

NOTE. All items normally supplied with the referenced flange are to be supplied with the adapters.

Part number M22641/21- (dash number from table I).

Supersession data See table II.

TABLE II. Supersession data.

Superseded type	Superseding part number M22641/21-
UG-1901/U	001
UG-1902/U	005
UG-1903/U	009
UG-1904/U	013

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